The Brent-Price Causal Effect of the 2026 Hormuz Disruption

A Synthetic Control Approach

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From the previous presentation

Original framing. *How does the topology of the global maritime oil trade network shape the dynamics of oil prices?*

- **Q1.** Can network topology improve oil-price forecasts beyond classical time-series baselines?
- **Q2.** How does a maritime disruption propagate into oil prices across horizons, and how long does its effect linger?

Why the pivot to causal SCM.

- We agreed to focus on **causal** estimation rather than forecasting — forecasting is hard and crowded; a credible counterfactual on a specific disruption is more tractable and policy-relevant.
- We reviewed candidate causal methods: event studies, DiD, network-shock propagation, synthetic control.
- **SCM** selected: cleanest combination of **inferential machinery** (in-space placebo) and **interpretable result** (donor-weighted counterfactual on the observed price), and it directly produces an ATT estimate rather than a shock decomposition.

Research question (Part 1)

What was the causal effect of the 2026 Strait of Hormuz disruption on the Brent crude price?

Estimand: average treatment effect on the treated (ATT) over a defined post-event window.

Approach

Synthetic control method (SCM) on the observed Brent price series. An ensemble of five model specifications spans the SCM design space.

Answer (today)

Brent traded at a **∼44%** premium above its donor-weighted counterfactual over the March–May 2026 post-event window.

Ensemble IQR across five models: [40%, 44%].

Part 2 (open) — *Possible Directions will be discussed at the end of today's presentation.*

How the field estimates oil-price impacts of disruptions

Structural VAR (Kilian 2009; Baumeister & Hamilton 2019)	**Event study** (Standard in energy & finance)	**Scenario models** (Dallas Fed WP 2609; OIES; Kiel PB 206; IEA OMR 2026)
Decomposes price into supply, demand, inventory shocks via structural restrictions. No counterfactual path. Restrictions contested for chokepoint events.	Measures abnormal return around T_0 using a pre-event benchmark. No path beyond a few days. Persistence not captured.	Assumes a supply-shortfall scenario; propagates via structural supply/demand model. Counterfactual is assumed, not read from observed prices.

Closest SCM-on-price precedents

Becker, Pfeifer & Schweikert 2021 (*Energy Economics*)	**Gharehgozli 2017** (*Economics Letters*)
SCM on Austrian retail fuel after a regulatory act. Cross-country donors. Reports 23% price reduction vs synthetic. Closest mechanical design to ours.	SCM on Iran GDP under sanctions 2011–15. Cross-country donors. Establishes SCM for geopolitical disruptions — but outcome is GDP, not a market price.

No published study applies SCM to a globally traded crude benchmark price under an exogenous geopolitical disruption. This is the gap we fill.

What we do differently, and what we add

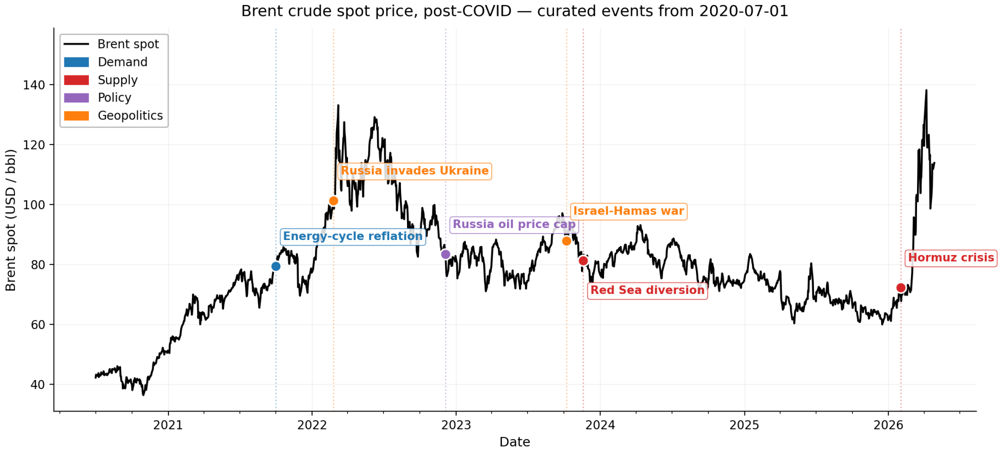
Three design extensions beyond the existing precedent

Retail fuel → crude benchmark	Cross-country → cross-asset donors	Regulatory → geopolitical treatment
Brent underpins retail fuel, CPI, and sovereign hedging.	No crude benchmark is Hormuz-clean. Donors: metals, FX, equities, rates.	Treatment is exogenous and unannounced. SUTVA is about asset-class routing, not policy endogeneity.

Two-event validation design Own contribution — no canonical SCM precedent

- Russia 2022 supplies a **magnitude-validation** case: the design recovers a credible magnitude on a known disruption before being applied to Hormuz.
- SCM counterfactual on Russia vs EIA STEO last pre-invasion forecast: $\Delta = \$3.98$/bbl ($\sim$5.7 pp on ATT) — independent-method agreement, no shared information path.

Brent spot price, post-COVID — curated event timeline



Brent spot, post-COVID. Events: demand (blue), supply (red), policy (purple), geopolitics (orange).

Methodology — ensemble of 5 models

Why an ensemble. Headline = median across models; IQR = model-uncertainty band.

Model	What it does	Reference
Convex SCM	Convex weights, no extrapolation	Abadie, Diamond & Hainmueller (2010, *JASA*)
Augmented SCM	Convex + ridge bias correction	Ben-Michael, Feller & Rothstein (2021, *JASA*)
Elastic-net	$L_1 + L_2$ regression, sparse donor weights	Doudchenko & Imbens (2016, NBER WP)
XGBoost	Gradient-boosted trees, flexible nonlinear	Chen & Guestrin (2016, *KDD*)
Bayesian Ridge	Bayesian linear with normal prior, donor PIPs	Brodersen et al. (2015)

Range covered. Convex (no extrap) → augmented (ridge fix) → regression (sparse) → nonlinear → Bayesian (uncertainty).

Model training and validation

	Pre-event (CV)		Treatment (post)	
Hormuz 2026	2024-06-03 → 2026-01-30	423 obs	2026-02-01 → 2026-05-01	∼59 obs
Russia 2022	2020-07-01 → 2022-02-23	421 obs	2022-02-24 → 2022-09-30	151 obs

Training. 5-fold expanding-window walk-forward CV (Hyndman & Athanasopoulos 2018; Bergmeir & Benítez 2012); hparams chosen by pooled val RMSE; final fit refits on full pre-event window.

Validation battery.

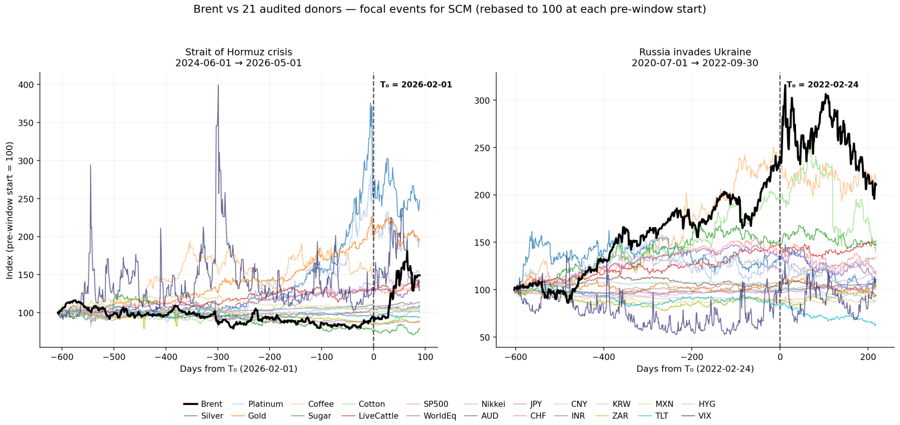
- **Model fit** — walk-forward CV val/train RMSE.
- **SUTVA** — qualitative donor audit on causal channels.
- **Inference** — in-space placebo, 21- and 33-donor pools.

Donor selection

21 audited donors (of 33 candidates). The full 33-donor catalog was audited for SUTVA and pre-event co-movement; 12 were dropped for event-specific exposure (Russia/Ukraine or Persian Gulf channels), leaving 21 in the shared pool used for both events — so we know the full scope of what was excluded and why. Inclusion requires *both*: **pre-treatment** co-movement with Brent through a common global factor, *and* **treatment** cleanliness (SUTVA).

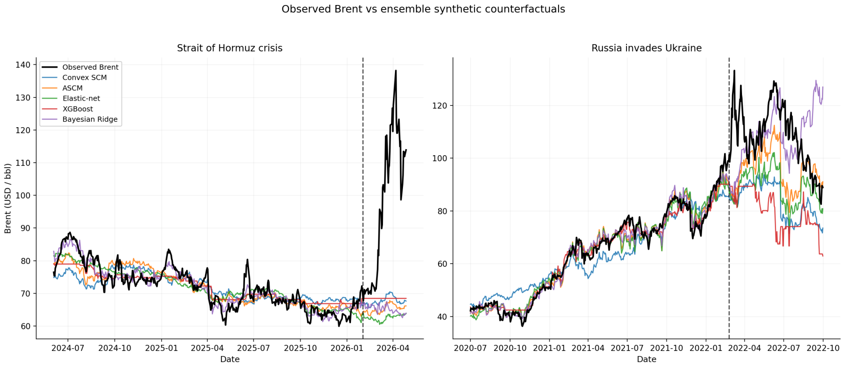
Category	Chosen donors	Factor channel (pre) & cleanliness (treatment)
Metals (3)	Silver, Platinum, Gold	Real rates / safe-haven / auto industry; no Russia or Gulf supply routing.
Agriculturals (4)	Coffee, Sugar, Cotton, LiveCattle	Global trade activity (Brazil / India / US producers); outside the Russia-Ukraine grain belt.
Equities (3)	SP500, WorldEq, Nikkei	Global growth / equity risk premium; no direct Russia or Gulf-energy exposure.
FX (8)	AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN	Safe-haven, EM growth, China demand; no petrocurrency or EU-energy-crisis loading.
Rates / credit (2)	TLT, HYG	US duration / credit-risk premia; not directly oil-linked.
Volatility (1)	VIX	S&P 500 implied vol — risk-on/off via equities.

Donor co-movement with Brent — audited pool



Brent (black) vs 21 audited donors, rebased to 100 at pre-window start; dashed line marks T_0. Same windows as the SCM.

Validation results — model fit



	Hormuz 2026			Russia 2022		
Model	**train RMSE**	**val RMSE**	**val/train**	**train RMSE**	**val RMSE**	**val/train**
Convex SCM	0.059	0.075	1.26	0.109	0.127	1.16
ASCM	0.047	0.064	1.35	0.043	0.088	**2.04**
Elastic-net	0.045	0.070	1.55	0.049	0.089	1.83
XGBoost	0.044	0.076	1.72	0.039	0.117	**3.00**
Bayesian Ridge	0.031	0.066	**2.10**	0.031	0.096	**3.07**

Validation results — in-space placebo

Procedure (Abadie 2010): refit the SCM treating each donor as treated. Rank Brent's post/pre RMSPE ratio against the N placebos (Brent + donors). **Permutation p-value = rank / N.** Rank 1 = Brent's break beats every placebo → strongest possible signal; floor $p = 1/N$ (0.045 at 21-donor, 0.029 at 33-donor).

Model	Hormuz 2026		Russia 2022	
	21-donor (N=22) rank → p	33-donor (N=34) rank → p	21-donor (N=22) rank → p	33-donor (N=34) rank → p
Convex SCM	**1 → 0.045**	**1 → 0.029**	12 → 0.545	17 → 0.500
ASCM	**1 → 0.045**	**1 → 0.029**	15 → 0.682	16 → 0.471
Elastic-net	**1 → 0.045**	**1 → 0.029**	9 → 0.409	14 → 0.412
XGBoost	**1 → 0.045**	**1 → 0.029**	**1 → 0.045**	5 → 0.147
Bayesian Ridge	**1 → 0.045**	**1 → 0.029**	14 → 0.636	18 → 0.529

- **Hormuz:** Brent ranks 1 in **every cell** → strongest possible signal across all 5 models and both pools.
- **Russia:** mid-pack (rank 9–18) for the linear models; only XGBoost at 21-donor reaches rank 1 (slips to 5 at 33-donor).

External validation — EIA STEO pre-invasion forecasts (Russia only)

Cross-method check. SCM counterfactual vs EIA's last pre-invasion Brent forecast (STEO Feb-22, issued 2022-02-08). No shared information path (SCM uses cross-assetco-movement; STEO uses fundamental supply-demand modelling)

Model	**Synth $/bbl**	**STEO Feb-22 $/bbl**	Δ
Convex SCM	83.55	84.94	−$1.39
ASCM	95.52	84.94	+$10.58
Elastic-net	88.92	84.94	+$3.98
XGBoost	79.28	84.94	−$5.66
Bayesian Ridge	107.91	84.94	+$22.97 (deg.)
Ensemble median	**88.92**	**84.94**	**+$3.98**

Implied ATT (Brent actual $108.54): SCM ensemble +22.1% vs STEO Feb-22 +27.8% ($\Delta \sim 5.7$ pp).

Headline result — ATT estimate

Why ATT? One treated unit (Brent) → effect *for that unit*, not a population ATE (Abadie 2010, 2021). Mean gap over the post-event window — integrates path-reversion, robust to single-day liquidity jumps.

Model	Hormuz 2026	Russia 2022
Convex SCM	+38.6%	+29.6%
ASCM	+43.7%	+13.3%
Elastic-net	+49.8%	+21.7%
XGBoost	+36.8%	+36.8%
Bayesian Ridge	+43.6%	+0.5% (deg.)
Ensemble median	+43.6%	+21.7%
IQR	[+38.6, +43.7]	[+13.3, +29.6]

Reading. Hormuz +43.6% with tight IQR — place against Dallas Fed / OIES / IEA scenarios. Russia +21.7% within ∼5.7 pp of STEO Feb-22 ATT (synth \$88.92 vs STEO \$84.94, Δ \$3.98/bbl); Bayesian Ridge degenerate, IQR excludes it.

Part 2 — future directions

Three directions build directly on the Brent SCM result established in Part 1.

1 LNG market causal estimation (JKM, TTF)

Instead of building the whole story around Brent, change the focus on commodity prices in general which were affected by the disruption. Apply the same SCM design to JKM Asian LNG spot price and/or TTF European LNG spot price. Identify the Hormuz LNG premium paid by Asian buyers vs a cross-asset counterfactual. First causal estimate of the Hormuz effect on Asian LNG pricing. Same methodology, different treated asset.

2 Local projections (LP) across horizons

Estimate the Brent treatment effect at $h = 1, 5, 14, 30$ days. Produces a horizon-by-horizon impulse response: how fast the effect builds and how long it persists. SCM gives the average ATT over a window; LP gives the *shape* of the path — different and complementary information.

3 Retail fuel and CPI pass-through

Estimate what share of the $\sim$44% Brent premium passed through to retail gasoline and headline CPI in oil-importing economies. Converts the upstream ATT into a consumer-price impact estimate. The Brent ATT is the upstream input; pass-through delivers the policy-relevant inflation number.

Currently, we are considering the following 3 options, however we're happy to discuss further options/suggestions.

Thank you

Questions and discussion

github.com/ElvisCasco/Brent_Analysis

Appendix — excluded donors

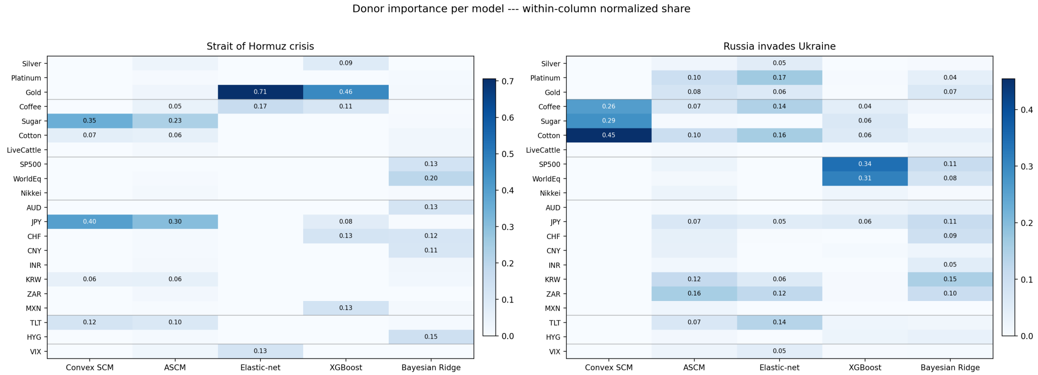
A. Categorical exclusions (never enter the candidate pool).

Category	Excluded & reason
Energy complex	WTI, NatGas, refined products (RBOB, heating oil, jet fuel), EIA inventories — mechanical SUTVA violation on any oil shock.
Petrocurrencies	CAD, NOK, BRL, RUB, COP, MYR — value partly set by oil exports.
FX unstable	TRY, ARS, EGP — domestic crisis dynamics dominate; noise, not signal.
Equities rejected	DAX, FTSE, TSX 60 (energy weight), KOSPI, Hang Seng (redundant with KRW / CNY).
Commodities rejected	Cocoa, OJ, Lean hogs (thin / idiosyncratic), Lumber (LBS$\rightarrow$LBR contract break), Aluminium (Rusal $\sim$6% supply).

B. Per-event SUTVA exclusions (33-donor pool $\rightarrow$ 21-donor strict-clean preferred).

- **Hormuz:** no H/M flags — audit clean on all 33 candidates. The 21-donor intersection is used to enable cross-event weight transfer (§5f), at the cost of 12 Hormuz-clean donors.
- **Russia H-tier** (direct supply/demand): Wheat, Corn, Palladium, EUR, DXY, BTC.
- **Russia M-tier** (terms-of-trade / secondary): Copper, IronOre, Soybeans, EM_Eq, GBP, US10Y.

Appendix — donor importance



Per-model raw metric: convex weight w_j (SCM, ASCM) / $|\hat{\beta}_j|$ (Elastic-net, Bayesian Ridge) / feature importance (XGBoost). Cells $= |w_j|$ normalised within each column to sum to 1.

Appendix — donor SUTVA cleanliness

Permutation mean-shift test at known T_0 on donor log-returns (Chow 1960 hypothesis + Lehmann & Romano 2005 permutation + Phipson & Smyth 2010 correction). Combined flag = permutation *or* Wilcoxon event-window rejects under Benjamini-Hochberg FDR ($\alpha = 0.10$).

	Permut.	**Event-wind.**	**Combined**	**Audit**
Hormuz 2026	0 / 33	1 / 33	1 / 33	0 / 33
Russia 2022	1 / 33	0 / 33	1 / 33	12 / 33

- **Hormuz: 32 / 33 agree.** Audit's "no Persian-Gulf routing" confirmed; only IronOre triggers (event-window).
- **Russia: 20 / 33 agree.** Audit's H/M tier captures *causal channels* (Wheat: Russia+Ukraine $\approx$ 30% world exports) that daily-return tests can't reproduce at $N = 33$ after FDR. Tests' lone rejection (CNY) is China zero-COVID, not Russia-treatment.

Appendix — in-time placebo & LOO

In-time placebo (Abadie 2015) — mean fake-period gap (%) at $T_0^{\text{fake}} = \text{real } T_0 - 6$ months, defaults hparams. **LOO** — range (max − min, pp) across leave-one-donor-out refits.

Model	In-time gap (%) Hormuz	In-time gap (%) Russia	LOO range (pp) Hormuz	LOO range (pp) Russia
Convex SCM	−11.9	+1.9	9.4	6.6
ASCM	−9.8	−4.2	8.9	13.4
Elastic-net	−10.6	+7.5	11.0	15.6
XGBoost	−12.4	+26.4	**4.7**	**4.9**
Bayesian Ridge	−12.2	−5.3	13.1	24.9 †

In-time. Hormuz −10 to −12% across all five models = pre-Hormuz runup donors did not absorb, not model failure. Russia fake window contains late-2021 oil rally + Jan-2022 invasion runup — partial real-effect contamination expected; XGBoost +26% low-power (training shrinks $\sim 31\%$).

LOO. Tightest: **XGBoost** (~ 5 pp, bold). Widest: **Bayesian Ridge Russia** († range crosses zero, −12.8 to +12.0) — LOO-fragile, consistent with its 3.07 walk-forward overfit ratio and 0.5% baseline. Ensemble median robust because Convex / ASCM / EN / XGBoost agree more closely.

Appendix — cross-event weight transfer

Cross-event weight transfer — a check on methodology generalization across events.
Russia-fitted weights applied to the Hormuz post-window vs independently-fit Hormuz gap:

Model	Independent	Transferred	Δ
Convex SCM	+38.6%	+47.0%	+8.4 pp
ASCM	+43.7%	+27.3%	−16.4 pp
Elastic-net	+49.8%	+23.2%	−26.6 pp
XGBoost	+36.8%	−58.2%	−95.0 pp (deg.)
Bayesian Ridge	+43.6%	−100.0%	−143.6 pp (deg.)

Reading.

- Convex SCM transfers cleanly (+8 pp drift).
- ASCM and Elastic-net give qualitatively consistent direction with magnitude attenuation.
- XGBoost and Bayesian Ridge degenerate — regime drift in Russia-fitted weights blows up the projection.
- Factor structure has shifted between 2020–22 and 2024–26, but the **sign and order of magnitude** of the Hormuz effect are robust under the linear-convex models.